

Tutorial 6

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Variational method with a parameter

Pick any ψ normalised

$$E_{gs} \leq \langle \psi | H | \psi \rangle \quad (1)$$

expected value.

Introduce parameter b in ψ .

$$\langle H \rangle = E_{gs}^{est}(b) \quad (2)$$

calculate $\frac{\partial \langle H \rangle}{\partial b} = 0$ find b^*

$$E_{gs}^{est}(b)^* \quad (3)$$

the highest bound for ground state.

Question 1

0.0.1 (I)

Given a particle trapped in an infinite well of size a that obeys the SE.

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi \quad (4)$$

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{a}\right) \quad (5)$$

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2ma^2} \quad (6)$$

General ansatz for ground state wavefunction

$$\psi(x) = \gamma(\beta) \left(x(a-x) + \frac{\beta}{a^2} (a-x)^2 \right) \quad (7)$$

(a)

An expression for $\gamma(\beta)$ to normalise the wave function

$$\int_0^a |\psi(x)|^2 dx = 1 \implies \gamma^2(\beta) \int_0^a \left[x(a-x) + \frac{\beta}{a^2} x^2 (a-x)^2 \right] dx = 1 \quad (8)$$

$$\implies \gamma(\beta) = \sqrt{\frac{630}{a^5} \frac{1}{\sqrt{b^2 + 9b + 21}}} \quad (9)$$

0.0.2 (b)

$$E(\beta) = \langle \psi | H | \psi \rangle = \int_0^a \psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) \right) dx \quad (10)$$

$$\psi'' = 2\gamma(\beta) \left(-1 + b - \frac{6bx}{a} + \frac{6bx^2}{a^2} \right) \quad (11)$$

$$E(\beta) = \frac{\hbar^2}{2m} \gamma(\beta) \left(\frac{a^3}{3} + \frac{2a^3b}{15} + \frac{2a^3b^2}{105} \right) \quad (12)$$

$$E(\beta) = \frac{3\hbar^2}{ma^2} \frac{2b^2 + 14b + 35}{b^2 + 9b + 21} \quad (13)$$

0.0.3 (C)

$$\frac{\partial E(\beta)}{\partial \beta} = 0 \implies \beta^* = \frac{1}{4} (\sqrt{133} - 7) \quad (14)$$

$$\langle H \rangle_{min} = E_{gs}^{est} = E(\beta^*) = \frac{\hbar^2}{ma^2} 2 (14 - \sqrt{133}) \quad (15)$$